FONTYS UNIVERSITY OF APPLIED SCIENCES
HOGESCHOOL INFORMATION AND COMMUNICATION TECHNOLOGY

DESIGN DOCUMENT

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Eindhoven, 2021

1. INTRODUCTION

1.1. TOPIC/CONCEPT

Application shows all places where there is a public chessboard. The main reason people will be using this app is to see different possible locations to meet up and play a match of chess. The goal of my Android project is an app to encourage people to meet up to play chess.

In the app you can leave your own chessboard targets to let other users know that there is a chessboard available.

1.2. GOAL OF THE PROJECT

The goal of the project is to improve my Android skill and by doing that create a prototype for the concept to show what I learned during this period.

1.3. SCOPE AND CONSTRAINTS

The scope of the project is to create a prototype that means It is just an early version and that the app will not have all functionality, only main functionality.

The constrains of this project would be time. The lack of time to build an app alone in a few weeks is not an easy task.

1.4. PROBLEM

Currently no one provides a solution for searching up locations where public chessboard could be found. Even Google Maps or other Map engines doesn’t search up any public chessboards.

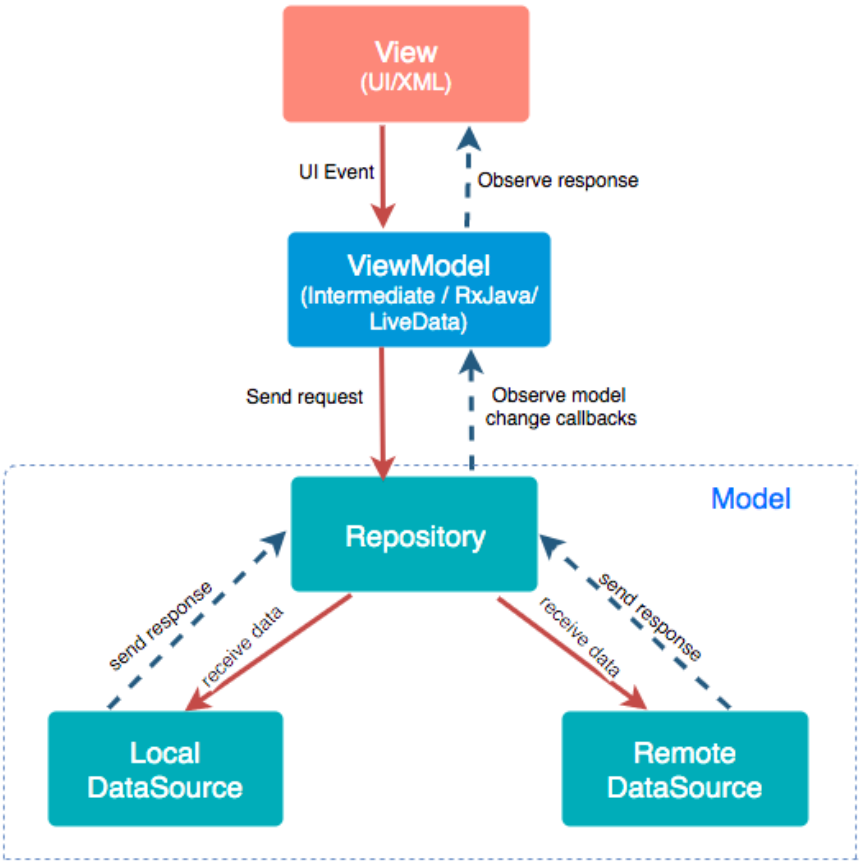
1.5. SOLUTION

To develop Chess Maps. So public chessboards would be more accessible for everyone.

2. TECHNICAL OWERVIEW

Technology	Description
Kotlin	Language in which the app was written in
MVVM	Model-View-ViewModel architecture used to develop the application
Firebase	Cloud database to store all the locations
Gitlabs	Version control platform
API	Google maps API and google places API where used.

2.1. ARCHITECTURE



3. UX/UI PRINCIPLES

3.1. MOBILE UX DESIGN KEY PRINCIPLES

1. **Cut out the clutter**

Don't overload users with information. Every added button, image or line will make the interface more complicated. A simple rule is one primary action per screen.

2. **Make navigation self-evident**

    1. Helping people navigate is a high priority of an app. If people can't find stuff your app is useless. Navigation must be coherent; you should use the right visuals so the navigation doesn't need explaining. Navigation must be consistent for the app; Don't move or hide navigation controls on different pages. Navigation should communicate the current location; the app should always indicate the current location.
3. **Create a seamless experience**
    1. You should not only think about mobile design but also how the app will look on different screens (laptop, tablet).
4. **Design finger-friendly tap-targets**
    1. Make sure tap-targets are big enough so they are easy to tap. Controls with a size of 7-10 mm can be accurately tapped with a finger. Also make sure there is anough space between tap targets.
5. **Text content should be legible**
    1. Text should be at least 11 points so it's legible at a typical viewing distance without zooming. Generous white space can make a page look more inviting and simple.
6. **Make interface elements clearly visible**
    1. Choose primary, secondary and accent colors for the app that support usability. Ensure enough color contrast so low vision users can use your app. Make sure there is enough contrast between text and background. Small text contrast ratio at least 4.5:1 against the background, large text at least 3:1. This applies on icons too.
7. **Design controls bases on hand position**
    1. Users only use one thumb for most of their phone. Place top level menu, frequently used controls, and common actions in the green zone. Place negative actions (delete or erase) in the red zone zo they can't be accidently tapped.
8. **Minimize need for typing**
    1. Typing is a slow and error-prone process. Remove unnecessary fields in forms. Use auto complete and personalized data.
9. **Test your design**
    1. Test the app with multiple users on different devices.

https://uxplanet.org/mobile-ux-design-key-principles-dee1a632f9e6

3.1.1. MOBILE UI DESIGN KEY PRINCILES

1. The Structure Principle
2. The architecture of the user interface should be purposeful and meaningful for the users.
3. The Simplicity Principle
4. Design should make tasks easy, communicate clearly in de users language and provide shortcuts.
5. The Visibility Principle
6. Design should make needed options and materials visible without distracting users with redundant information. Don't overwhelm and confuse users.

7. The Feedback Principle
8. Design should keep users informed about what is going on in a way the user understands.
9. The Tolerance Principle
10. Design should be flexible and tolerant. Allow mistakes by allowing undoing while preventing errors where is possible.
11. The Reuse Principle
12. Design should remember behavior from users to reuse. This way the user has to remember and think of less.

https://careerfoundry.com/en/blog/ui-design/how-to-design-a-mobile-app-using-user-interface-design-principles/

4. USER STUDY

4.1. TARGET AUDIENCE

Target users for this all application varies a lot. It could be anyone who plays chess. A lot of elderly people like to play chess in public parks, this could be a great all to discover new places they could play.

4.2. GEOGRAPHICS

Anyone in the world could participate in this application and add as many chessboards available.

4.3. DEMOGRAPHICS

- Adults 20-50 years old. People of this age who enjoy playing chess or being in the park.
- Elder people who have an interest in chess and would like to discover new places to play.
- Youngsters 9 – 16 years old. A lot of parents encourage their children to play chess and surprisingly they can be very good at it. Maybe this application would encourage them to play chess even after chess class after school is over.

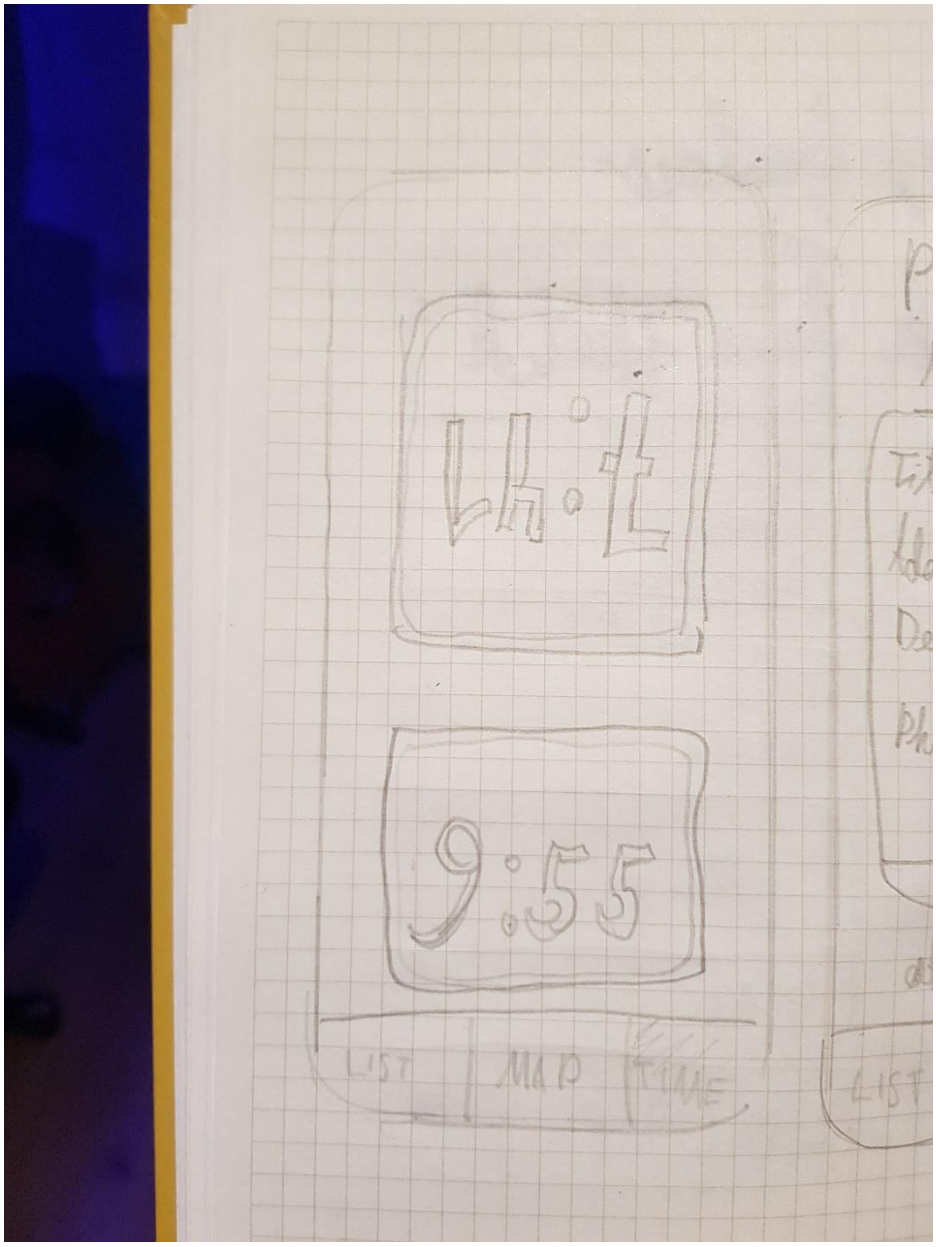
4.4. PERSONA

Name: Peter Wayne
Age: 56

Occupation: Retired
City: Eindhoven
Goals and motivations: Peter is retired and he likes to spend his time in the park. In the park there is a public chess board that he likes to challenge everyone to play versus him. Peter likes to spend time like this and he would like to find new places like that so he could make some new friends with the similar hobbies.
Problems faced now:

- Doesn't know any other locations where he can find a public chessboard.

5. SKETCHES



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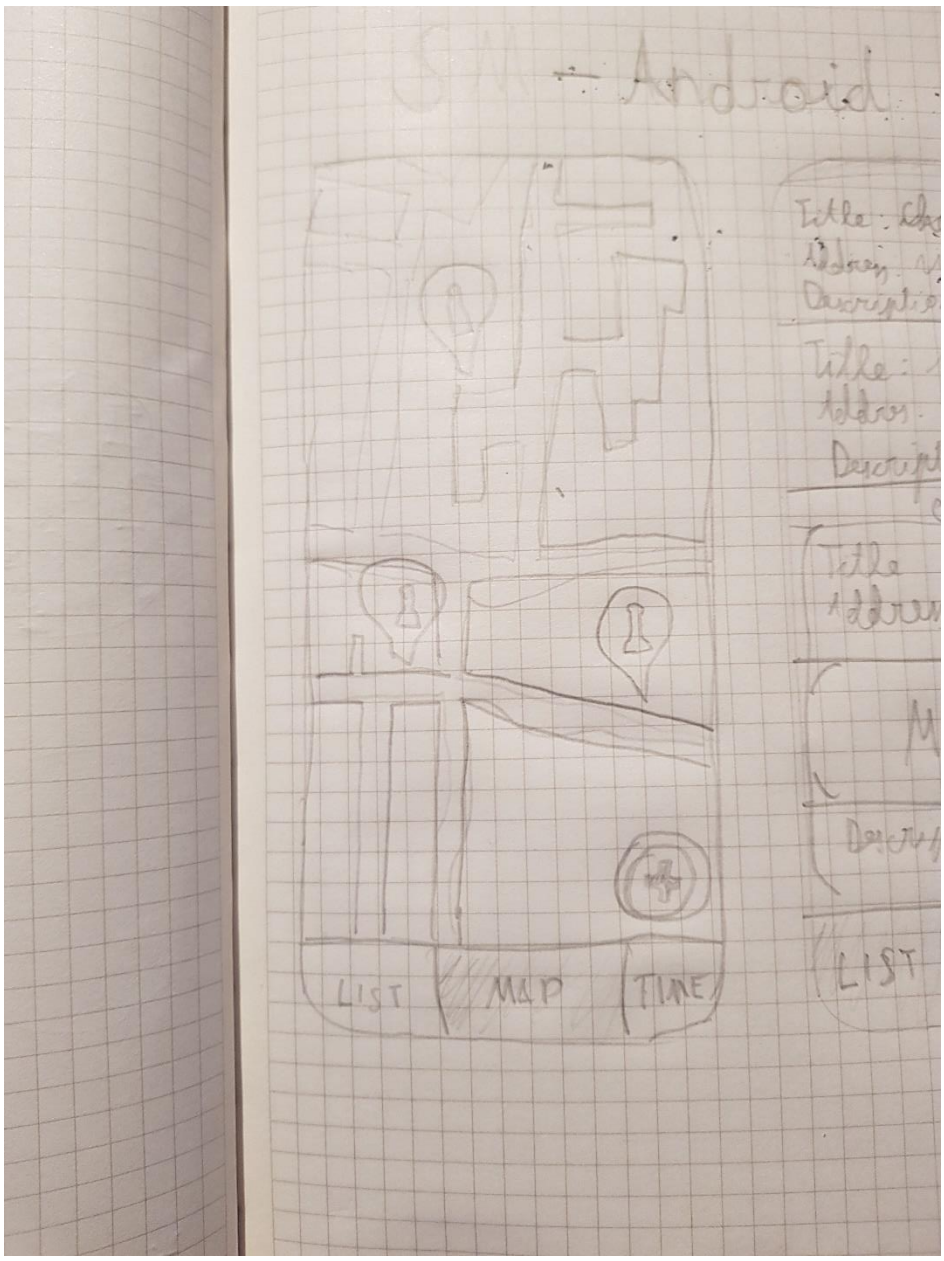
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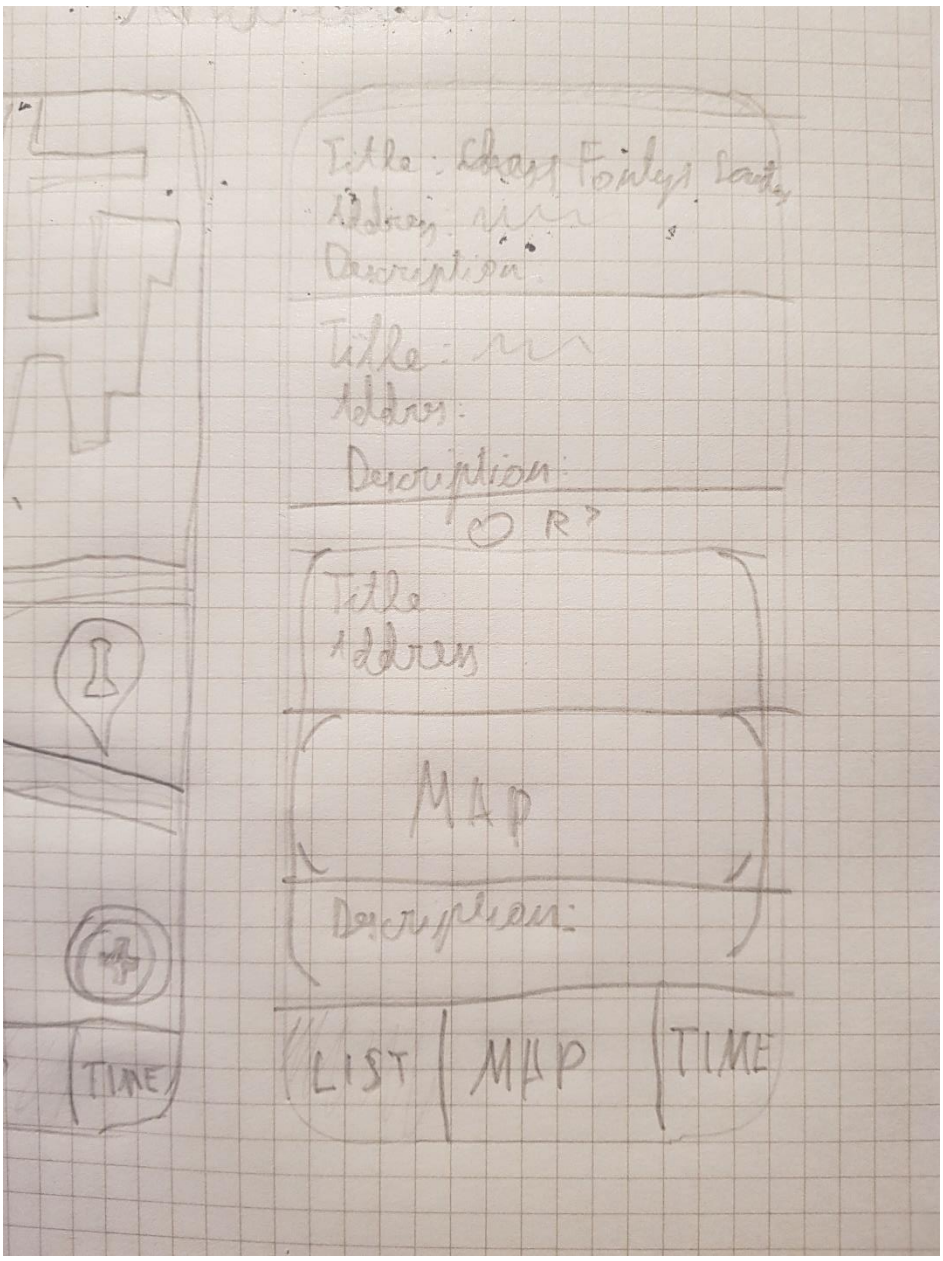
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